KCE PATHWAYS · GRADE 3 · UNIT 6: BUILDING TOGETHER · WEEK 26 · MATH

The Bike Repair Shop in Portland — Teacher Lesson Plan & Answer Key

Learning objectives: Students multiply a one-digit number by a multiple of 10 by thinking in tens (4 × 30 = 12 tens = 120), use the commutative property to switch factors and the associative property to group three factors, use the distributive property to break a factor into a 10 and some ones (4 × 18 = 4 × 10 + 4 × 8), and choose a smart strategy for two-step problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce Ms. Castellanos's bicycle repair shop in Portland, Oregon, with her helpers Cassius, Celeste and Cayden.
2. Warm-Up: Unit 5 Review (10 pts): two-step problems, doing × or ÷ before + or −, and rounding to check. If a student misses both, review "find the hidden question first, then do the × or ÷ part first."
3. Picture It: The Bike Repair Shop in Portland. Spokes and patches come in boxes of 10, 20 or more, so the helpers multiply by multiples of 10 (Portland has 12 crossings over the Willamette River, so people call it Bridgetown). Ask: why is it handy to think in tens at a repair shop?
4. Word Wrangler: 6 vocab cards (multiple of 10, factor, product, commutative property, associative property, distributive property).
5. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W26.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
6. Multiplying by Multiples of 10 (10 pts): use a known fact, then think in tens (6 × 80 = 48 tens = 480). Tip: the fact tells how many tens; watch for answers with one zero too many or too few.
7. Switch and Group: Commutative and Associative Properties (10 pts): naming the property, grouping three factors ((2 × 4) × 30 = 2 × (4 × 30)) and a find-the-mistake item (he added instead of multiplying). Tip: the parentheses show which part to multiply first.
8. Break It Apart: The Distributive Property (10 pts): break a factor into a 10 and the ones that are left, multiply both parts, then add (6 × 14 = 60 + 24 = 84). Watch for the common slip of multiplying only one part.
9. Smart Strategies for Two-Step Problems (10 pts): money problems and products that share a factor (3 × 40 + 3 × 50 = 3 × 90). Tip: look for a pair that makes 10 or 100, like 2 × 50.
10. Repair Challenge: Multiply with Tens (10 pts): two-step and three-step problems, the break-apart trick (4 × 19, 7 × 13) and a missing number (6 × 40 = 8 × 30). Ask the student to name step 1 before solving.
11. Bonus Practice: The Parts Wall (XP only, not scored).

12. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches the properties and vocabulary to examples and meanings, Set 2 matches multiples-of-10 facts to their products, Set 3 matches products to their break-apart equations.
13. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters for names of people and places and the title Ms., end mark, list commas, and one verb fix); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.
14. Speed Arena and Keyboard Corral (XP only, not scored).
15. Boss Battle, Part 1: The Saturday Rush (10 pts, 5 each): two-step money problems, grouping three factors, the break-apart trick, and which-explanation or find-the-mistake items (5 × 60 is 300, not 3,000; multiply both parts).
16. Boss Battle, Part 2: Closing Time (10 pts, 5 each): missing numbers and factors, a two-step spokes or reflectors problem, and find-the-mistake items where the student types the correct product (for example, 8 × 40 = 320, not 32).
17. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W26.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
18. Results (Repair Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Repair Shop Hero!” with the congratulations message “Repair shop hero! You kept every bike rolling!” and a big confetti burst; below 85 shows “Great Effort Today!” with encouragement (“Every mechanic gets better with practice.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: UNIT 5 REVIEW (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Two steps! Ms. Castellanos had 85 bike bells. She sold 3 boxes with 9 bells in each box. How many bells does she have left?	**58** (also accepted: 58 bells, 58 bike bells)	Step 1: 3 × 9 = 27 bells sold. Step 2: 85 − 27 = 58 bells left.
w2_s0	Set 1: Which explanation is correct for 6 + 4 × 5? [Multiply first: 4 × 5 = 20. Then 6 + 20 = 26. / Add first: 6 + 4 = 10. Then 10 × 5 = 50. / Work from right to left: 5 × 4 = 20, and 20 − 6 = 14. / Add all three numbers: 6 + 4 + 5 = 15.]	**Multiply first: 4 × 5 = 20. Then 6 + 20 = 26.**	When an equation has × and +, do the × part first. 4 × 5 = 20, and 6 + 20 = 26.
w1_s1	Set 2: Cassius estimates 389 + 207 by rounding each number to the nearest hundred. What is his estimate?	**600**	389 rounds to 400, and 207 rounds to 200. 400 + 200 = 600. The exact sum, 596, is close to 600.
w2_s1	Set 2: Find the mistake. Celeste solved 30 − 3 × 6 = n and got n = 162. What went wrong? [She subtracted first. Multiply first: 3 × 6 = 18. Then 30 − 18 = 12, so n = 12. / Nothing is wrong. n = 162. / She should add: 30 + 3 + 6 = 39. / She should divide: 30 ÷ 3 = 10, so n = 10.]	**She subtracted first. Multiply first: 3 × 6 = 18. Then 30 − 18 = 12, so n = 12.**	Do the × part before the − part. 3 × 6 = 18, and 30 − 18 = 12.
w1_s2	Set 3: Two steps! Cayden has 7 boxes of tire patches with 6 patches in each box. He uses 15 patches. How many patches are left?	**27** (also accepted: 27 patches, 27 tire patches)	Step 1: 7 × 6 = 42 patches. Step 2: 42 − 15 = 27 patches left.

ID	Question	Answer	Why (explain to student)
w2_s2	Set 3: Solve the equation: 8 × 3 + n = 30. What is n?	**6** (also accepted: n = 6, n=6)	Step 1: 8 × 3 = 24. Step 2: 24 + n = 30, so n = 30 − 24 = 6.

MULTIPLYING BY MULTIPLES OF 10 (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
a1_s0	Set 1: A box holds 40 bike spokes. Celeste opens 7 boxes. How many spokes does she have?	**280** (also accepted: 280 spokes, 280 bike spokes)	7 × 40 = 7 × 4 tens = 28 tens = 280 spokes.
a2_s0	Set 1: What is 5 × 80? Think: 5 × 8 tens.	**400**	5 × 8 = 40, so 5 × 8 tens = 40 tens = 400. Watch out: 40 tens is 400, not 40.
a1_s1	Set 2: Cayden stacks 8 packs of inner tubes. Each pack has 30 inner tubes. How many inner tubes is that?	**240** (also accepted: 240 inner tubes, 240 tubes)	8 × 30 = 8 × 3 tens = 24 tens = 240 inner tubes.
a2_s1	Set 2: Find the mistake. Celeste says 6 × 50 = 3,000. What went wrong? [6 × 5 tens = 30 tens, and 30 tens is 300. She wrote one zero too many. / Nothing is wrong. 6 × 50 = 3,000. / 6 × 50 should be 56, because you put the 6 next to the 50. / 6 × 50 should be 30, because 6 × 5 = 30.]	**6 × 5 tens = 30 tens, and 30 tens is 300. She wrote one zero too many.**	6 × 50 means 6 groups of 5 tens. That is 30 tens, and 30 tens = 300.
a1_s2	Set 3: Cassius pumps up 9 tires. Each tire needs 20 pumps of air. How many pumps does he do in all?	**180** (also accepted: 180 pumps)	9 × 20 = 9 × 2 tens = 18 tens = 180 pumps.
a2_s2	Set 3: Find the missing factor: 3 × ___ = 210.	**70**	3 × 7 = 21, so 3 × 7 tens = 21 tens = 210. The missing factor is 70.

SWITCH AND GROUP: COMMUTATIVE AND ASSOCIATIVE PROPERTIES (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
m1_s0	Set 1: Which property does this show? 5 × 70 = 70 × 5 [The commutative property / The associative property / The distributive property / It is not true, because the order matters.]	**The commutative property**	The two factors switched places, and the product stays the same. That is the commutative property.
m2_s0	Set 1: Cayden packs 2 crates. Each crate has 3 boxes, and each box holds 50 bolts. How many bolts in all? Hint: 2 × 3 × 50.	**300** (also accepted: 300 bolts)	Group the first two factors: (2 × 3) × 50 = 6 × 50 = 30 tens = 300 bolts.
m1_s1	Set 2: Celeste knows (3 × 2) × 40 = 3 × (2 × 40). What is the product?	**240**	(3 × 2) × 40 = 6 × 40 = 240. Or 3 × (2 × 40) = 3 × 80 = 240. Same product!
m2_s1	Set 2: Which explanation is correct? Why does 7 × 30 = 210? [30 = 3 × 10, so 7 × 30 = (7 × 3) × 10 = 21 × 10 = 210. / Add a zero to each number: 70 + 30 = 100, and 100 + 110 = 210. / 7 + 30 = 37, and 37 rounds to 210. / Because 7 × 3 = 210.]	**30 = 3 × 10, so 7 × 30 = (7 × 3) × 10 = 21 × 10 = 210.**	Write 30 as 3 × 10 and regroup with the associative property: (7 × 3) × 10 = 21 × 10 = 210.
m1_s2	Set 3: Multiplying 5 × 2 first makes 5 × 6 × 2 easier. What is the product?	**60**	Switch and group: 5 × 2 = 10, then 10 × 6 = 60.

ID	Question	Answer	Why (explain to student)
m2_s2	Set 3: Find the mistake. Cassius wrote 4 × 3 × 20: "4 × 3 = 12, and 12 + 20 = 32." What went wrong? [He added instead of multiplying. 12 × 20 = 240, or 4 × (3 × 20) = 4 × 60 = 240. / Nothing is wrong. The product is 32. / He should start with 3 + 20 = 23. / He should subtract: 20 − 12 = 8.]	**He added instead of multiplying. 12 × 20 = 240, or 4 × (3 × 20) = 4 × 60 = 240.**	Every sign is ×, so keep multiplying: 4 × 3 = 12, and 12 × 20 = 240.

BREAK IT APART: THE DISTRIBUTIVE PROPERTY (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
d1_s0	Set 1: Celeste breaks apart 6 × 14 into 6 × 10 + 6 × 4. What is 6 × 14?	**84**	6 × 10 = 60 and 6 × 4 = 24. 60 + 24 = 84.
d2_s0	Set 1: A rectangle is 7 rows tall. It is split into a part 5 columns wide and a part 3 columns wide. Which equation matches? [7 × (5 + 3) = 7 × 5 + 7 × 3 / 7 × (5 + 3) = 7 × 5 + 3 / 7 × (5 + 3) = 7 + 5 + 3 / 7 × (5 + 3) = 7 × 5 × 3]	**7 × (5 + 3) = 7 × 5 + 7 × 3**	Both parts have 7 rows, so multiply 7 by each part: 7 × 5 = 35 and 7 × 3 = 21. 35 + 21 = 56 = 7 × 8.
d1_s1	Set 2: Use the break-apart trick: 3 × 17 = 3 × 10 + 3 × 7. What is 3 × 17?	**51**	3 × 10 = 30 and 3 × 7 = 21. 30 + 21 = 51.
d2_s1	Set 2: Find the mistake. Cayden says 5 × 16 = 5 × 10 + 6 = 56. What went wrong? [He forgot to multiply the 6 by 5. 5 × 10 + 5 × 6 = 50 + 30 = 80. / Nothing is wrong. 5 × 16 = 56. / He should add 5 + 16 = 21. / He should break 16 into 1 and 6, so 5 × 16 = 5 + 30 = 35.]	**He forgot to multiply the 6 by 5. 5 × 10 + 5 × 6 = 50 + 30 = 80.**	Both parts get multiplied: 5 × 10 = 50 and 5 × 6 = 30. 50 + 30 = 80.
d1_s2	Set 3: Cassius breaks apart 8 × 12 into 8 × 10 + 8 × 2. What is 8 × 12?	**96**	8 × 10 = 80 and 8 × 2 = 16. 80 + 16 = 96.
d2_s2	Set 3: Find the missing number: 4 × 23 = 4 × 20 + 4 × ___	**3**	23 = 20 + 3, so 4 × 23 = 4 × 20 + 4 × 3 = 80 + 12 = 92.

SMART STRATEGIES FOR TWO-STEP PROBLEMS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: Two steps! Cayden sells 4 bike helmets for $30 each and 2 bike lights for $20 each. How many dollars is that in all?	**160** (also accepted: $160, 160 dollars, $160.00, $ 160)	Step 1: 4 × $30 = $120 and 2 × $20 = $40. Step 2: $120 + $40 = $160.
e2_s0	Set 1: Find the missing number: 3 × 40 + 3 × 50 = 3 × ___	**90**	Both products have 3 groups, so put the groups together: 40 + 50 = 90. 3 × 90 = 270, and 120 + 150 = 270.
e1_s1	Set 2: Two steps! A tune-up costs $40. On Saturday the shop does 7 tune-ups. Customers also pay $50 in all for new bells. How many dollars does the shop get?	**330** (also accepted: $330, 330 dollars, $330.00, $ 330)	Step 1: 7 × $40 = $280. Step 2: $280 + $50 = $330.
e2_s1	Set 2: Which is the easiest correct way to find 2 × 9 × 50? [Switch and group: 2 × 50 = 100, then 100 × 9 = 900. / 2 × 9 = 18, then 18 + 50 = 68. / 9 × 50 = 45, then 2 × 45 = 90. / 2 + 9 + 50 = 61.]	**Switch and group: 2 × 50 = 100, then 100 × 9 = 900.**	2 × 50 = 100 is easy, and 9 groups of 100 is 900. The product is 900 no matter how you group the factors.

ID	Question	Answer	Why (explain to student)
e1_s2	Set 3: Two steps! Ms. Castellanos had 500 spokes. She used 6 boxes with 70 spokes in each box. How many spokes are left?	**80** (also accepted: 80 spokes)	Step 1: 6 × 70 = 42 tens = 420 spokes used. Step 2: 500 − 420 = 80 spokes left.
e2_s2	Set 3: Cassius sees 8 × 30 + 2 × 30. He says it is the same as 10 × 30. What is the total?	**300**	8 groups of 30 plus 2 groups of 30 is 10 groups of 30. 10 × 30 = 300. Check: 240 + 60 = 300.

REPAIR CHALLENGE: MULTIPLY WITH TENS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: Two steps! Celeste orders 4 boxes of reflectors with 60 reflectors in each box. She puts 90 reflectors on bikes. How many reflectors are left?	**150** (also accepted: 150 reflectors)	Step 1: 4 × 60 = 24 tens = 240 reflectors. Step 2: 240 − 90 = 150 reflectors left.
c2_s0	Set 1: Use the break-apart trick to find 4 × 19.	**76**	4 × 19 = 4 × 10 + 4 × 9 = 40 + 36 = 76.
c1_s1	Set 2: Cayden test-rides each repaired bike for 20 minutes. He test-rides 3 bikes on Friday and 5 bikes on Saturday. How many minutes does he ride in all?	**160** (also accepted: 160 minutes)	Step 1: 3 + 5 = 8 bikes. Step 2: 8 × 20 = 160 minutes. Or 60 + 100 = 160.
c2_s1	Set 2: Use the break-apart trick to find 7 × 13.	**91**	7 × 13 = 7 × 10 + 7 × 3 = 70 + 21 = 91.
c1_s2	Set 3: Water bottles come in cases of 30. The shop gets 3 cases in the morning and 4 cases in the afternoon. Then it sells 50 bottles. How many bottles are left?	**160** (also accepted: 160 bottles, 160 water bottles)	Step 1: 3 + 4 = 7 cases. Step 2: 7 × 30 = 210 bottles. Step 3: 210 − 50 = 160 bottles left.
c2_s2	Set 3: Find the missing number: 6 × 40 = 8 × ___	**30**	6 × 40 = 240. Which number times 8 makes 240? 8 × 3 tens = 24 tens = 240, so the missing number is 30.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match commutative property	**3 × 40 = 40 × 3**	2 points if matched on the first try.
match_s0_2	Set 1: Match associative property	**(2 × 5) × 6 = 2 × (5 × 6)**	2 points if matched on the first try.
match_s0_3	Set 1: Match distributive property	**4 × 12 = 4 × 10 + 4 × 2**	2 points if matched on the first try.
match_s0_4	Set 1: Match multiple of 10	**a number like 30, 60, or 90**	2 points if matched on the first try.
match_s0_5	Set 1: Match product	**the answer to a multiplication problem**	2 points if matched on the first try.
match_s1_1	Set 2: Match 3 × 70	**210**	2 points if matched on the first try.
match_s1_2	Set 2: Match 8 × 50	**400**	2 points if matched on the first try.
match_s1_3	Set 2: Match 6 × 60	**360**	2 points if matched on the first try.
match_s1_4	Set 2: Match 9 × 30	**270**	2 points if matched on the first try.
match_s1_5	Set 2: Match 5 × 20	**100**	2 points if matched on the first try.
match_s2_1	Set 3: Match 6 × 15	**6 × 10 + 6 × 5**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s2_2	Set 3: Match 4 × 16	**4 × 10 + 4 × 6**	2 points if matched on the first try.
match_s2_3	Set 3: Match 3 × 19	**3 × 10 + 3 × 9**	2 points if matched on the first try.
match_s2_4	Set 3: Match 8 × 13	**8 × 10 + 8 × 3**	2 points if matched on the first try.
match_s2_5	Set 3: Match 5 × 17	**5 × 10 + 5 × 7**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: portland is in oregon	**Portland is in Oregon.**	Portland is the name of a city and Oregon is the name of a state, so both get capital letters. End the telling sentence with a period.
fs2_s0	Set 1: Fix the sentence: we fixed tires chains and brakes	**We fixed tires, chains, and brakes.** (also accepted: We fixed tires, chains and brakes.)	Start with a capital W, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: how many spokes are in the box	**How many spokes are in the box?**	Start with a capital H. This sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: cassius celeste and cayden oiled the chains	**Cassius, Celeste, and Cayden oiled the chains.** (also accepted: Cassius, Celeste and Cayden oiled the chains.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: did ms. castellanos ride across tilikum crossing	**Did Ms. Castellanos ride across Tilikum Crossing?**	Start with a capital D. Ms. Castellanos is a name, and Tilikum Crossing is the name of a bridge, so they get capital letters. The sentence asks something, so it ends with a question mark.
fs2_s2	Set 3: Fix the sentence: the bikes is ready for their riders	**The bikes are ready for their riders.**	Start with a capital T and end with a period. Bikes means more than one, so use are, not is.

BOSS BATTLE: THE SATURDAY RUSH (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Two steps! Ms. Castellanos sells 6 bike locks for $30 each. Then a customer returns 1 lock and gets the money back. How many dollars does she keep?	**150** (also accepted: $150, 150 dollars, $150.00, $ 150)	Step 1: 6 × $30 = $180. Step 2: $180 − $30 = $150. Or think: 5 locks are kept, and 5 × $30 = $150.
bb2_s0	Set 1: Which explanation is correct? Celeste says 5 × 60 = 300. Cayden says 5 × 60 = 3,000. [Celeste is right. 5 × 6 tens = 30 tens, and 30 tens = 300. / Cayden is right, because 5 × 6 = 30 and you add two zeros. / They are both wrong. 5 × 60 = 65. / They are both right, because 300 and 3,000 are close.]	**Celeste is right. 5 × 6 tens = 30 tens, and 30 tens = 300.**	5 × 60 is 5 groups of 6 tens, which is 30 tens. 30 tens = 300.

ID	Question	Answer	Why (explain to student)
bb1_s1	Set 2: The shop has 2 shelves. Each shelf has 4 boxes, and each box holds 20 bells. How many bells are there?	**160** (also accepted: 160 bells)	Group the factors: (2 × 4) × 20 = 8 × 20 = 160 bells.
bb2_s1	Set 2: Use the break-apart trick to find 6 × 17.	**102**	6 × 17 = 6 × 10 + 6 × 7 = 60 + 42 = 102.
bb1_s2	Set 3: Two steps! A tune-up costs $50. The shop does 4 tune-ups in the morning and 3 in the afternoon. How many dollars is that?	**350** (also accepted: $350, 350 dollars, $350.00, $ 350)	Step 1: 4 + 3 = 7 tune-ups. Step 2: 7 × $50 = $350. Or $200 + $150 = $350.
bb2_s2	Set 3: Find the mistake. Cayden says 3 × 26 = 3 × 20 + 6 = 66. What went wrong? [He forgot to multiply the 6 by 3. 3 × 20 + 3 × 6 = 60 + 18 = 78. / Nothing is wrong. 3 × 26 = 66. / He should add: 3 + 26 = 29. / He should find 3 × 2 = 6 and 3 × 6 = 18, so 3 × 26 = 6 + 18 = 24.]	**He forgot to multiply the 6 by 3. 3 × 20 + 3 × 6 = 60 + 18 = 78.**	Break 26 into 20 + 6 and multiply both parts by 3: 3 × 20 = 60 and 3 × 6 = 18. 60 + 18 = 78. The 2 in 26 means 2 tens, so 3 × 2 tens = 6 tens = 60.

BOSS BATTLE: CLOSING TIME (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: Find the missing number: 7 × 80 = 7 × 8 × ___	**10**	80 = 8 × 10, so 7 × 80 = 7 × 8 × 10 = 56 × 10 = 560.
bb4_s0	Set 1: Find the mistake. Cassius says 9 × 14 = 9 × 10 + 4 = 94. Type the correct product.	**126**	Multiply both parts: 9 × 10 = 90 and 9 × 4 = 36. 90 + 36 = 126.
bb3_s1	Set 2: Find the missing factor: ___ × 90 = 450.	**5**	5 × 9 = 45, so 5 × 9 tens = 45 tens = 450. The missing factor is 5.
bb4_s1	Set 2: Celeste writes 4 × 50 + 4 × 30 = 4 × 80. What is the total?	**320**	4 × 80 = 32 tens = 320. Check: 200 + 120 = 320.
bb3_s2	Set 3: Two steps! The shop had 400 reflectors. It used 5 packs with 60 reflectors in each pack. How many reflectors are left?	**100** (also accepted: 100 reflectors)	Step 1: 5 × 60 = 30 tens = 300 reflectors used. Step 2: 400 − 300 = 100 left.
bb4_s2	Set 3: Find the mistake. Celeste says 8 × 40 = 32. Type the correct product.	**320**	8 × 4 = 32, so 8 × 4 tens = 32 tens. 32 tens = 320, not 32.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W26.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: 50 bike bells; sells 4 boxes with 6 bells in each box	26 bells left. Step 1: 4 × 6 = 24 bells sold. Step 2: 50 − 24 = 26.
2	Fill in: 5 × 10 = ___ and 5 × 3 tens = ___ tens = ___	5 × 10 = 50. 5 × 3 tens = 15 tens = 150.
3	Which is the same as 4 × 7? (7 × 4, 4 + 7, 47)	7 × 4. The factors can switch places and the product stays the same (commutative property).
4	Draw 2 groups with 3 tens in each group	6 tens in all, which is 60. Check that the drawing shows 2 groups of 3 tens.

#	Prompt	Model answer / grading note
5	Find 6 × 2 × 5 (try 2 × 5 first)	60. 2 × 5 = 10, and 6 × 10 = 60. Accept any reasonable attempt before the lesson.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	6 boxes with 40 spokes in each box; show how you used tens	240 spokes. 6 × 4 = 24, so 6 × 4 tens = 24 tens = 240.
2	500 tire patches; uses 7 packs with 60 patches in each pack	Step 1: 7 × 60 = 42 tens = 420 patches used. Step 2: 500 − 420 = 80 patches left.
3	Cayden says 8 × 50 = 40	He forgot that 50 is 5 tens. 8 × 5 tens = 40 tens, and 40 tens = 400. The correct product is 400.
4	Break apart 7 × 14 into 7 × 10 + 7 × 4; area model	98. The area model has 7 rows split into a 10-wide part (7 × 10 = 70) and a 4-wide part (7 × 4 = 28). 70 + 28 = 98.
5	Which property does (3 × 2) × 40 = 3 × (2 × 40) show?	associative (the grouping changes; both sides equal 240).

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Find the missing number: 9 × 40 = 6 × ___	60. 9 × 40 = 36 tens = 360, and 6 × 6 tens = 36 tens = 360, so 6 × 60 = 360.